IA#1 — the system you are specifying for: table-order

CSC13114 · 23KTPM1 · read this before you write your spec
table-order is a small ordering system for one restaurant. It is already in production. You have not built it, and you do not need to — your spec adds one feature to it, **split the bill at a table**. Everything below is how the system works **today**. Anything not written here, your spec has to decide.

Who uses it

- **Customers** scan a QR code on their table. The browser gets a *guest session* (a cookie, no account, no login) tied to that table. Several phones at one table means several guest sessions on the same table.
- **Waiters** use a staff app, logged in with a staff account (JWT, role waiter). They can see every table, add items, apply a discount, and close an order.
- The **kitchen** has a screen that shows order items. It is not affected by payment in any way.

Data, as it is now (PostgreSQL)

Table	Columns that matter
tables	id, code (printed in the QR, e.g. T07), status — free | occupied
orders	id, table_id, status — open | paid | closed, discount_percent (0–50, set by a waiter), opened_at
order_items	id, order_id, menu_item_id, qty, unit_price, added_by — guest session id or staff id, added_at
payments	id, order_id, amount, status — pending | succeeded | failed, provider_ref, created_at

- A table has **at most one open order** at a time.
- Prices are **integers in đồng**, VAT (8%) already included. There are no fractional đồng anywhere in the database.
- The order total is computed, not stored: sum(qty × unit_price), then the discount, rounded to the nearest đồng.

API, as it is now (Express, JSON)

Method and path	Who	What it does
GET /api/tables/:code/order	guest at that table, staff	the open order with its items and

Method and path	Who	What it does
		total
POST /api/tables/:code/order/items	guest at that table, staff	adds items; opens an order if none is open
POST /api/orders/:id/discount	staff	sets discount_percent
POST /api/orders/:id/pay	guest at that table	creates a pending payment for **the whole total** and returns a payment-page URL
POST /api/webhooks/payment	the payment provider	reports succeeded or failed for a provider_ref
POST /api/orders/:id/close	staff	closes a paid order and frees the table

Errors today have one shape: { "error": "<code>", "message": "<text>" }.

The payment provider

- Payment happens on the provider's own page (cards and e-wallets). Our server never sees card details.
- The result arrives later, through the webhook. The webhook is signed, may arrive **more than once** for the same payment, may arrive **minutes late**, and occasionally never arrives — a payment can stay pending.
- Refunds exist in the provider's API but table-order does not use them yet.

Live updates

Every phone at a table and the staff app hold a WebSocket. When the order changes (items added, discount, payment status), the server broadcasts the new order to that table and to staff.

What you are adding

The people at a table want to **pay separately, each from their own phone**, instead of one person paying the whole bill. The kitchen is not affected. The order stays open until the whole amount is paid.
That sentence is all the product owner wrote. Your spec turns it into something another developer — or an agent — could build without asking you anything.